Muhamad Rafli Parindang

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Education

Institut Teknologi Bandung (ITB) — Bandung, Indonesia
Fast-Track B.S. & M.S. in Mechanical Engineering (PISM ITB) | GPA: 3.62 / 4.00 — *Aug. 2023 – Present*

- **Master's Specialization**: Program Penyatuan Sarjana Magister (PISM) with research focus on **Combustion**.
- **Relevant Coursework**: Fluid Mechanics, Thermodynamics, Heat & Mass Transfer, Mechanical Design & Manufacturing, Mechanics of Materials, Piping System Design, Mechatronics, Numerical Methods.

Experience

PT Pindad (Persero) — Bandung, Indonesia
Mechanical Engineering Intern – Innovation & Business Development Division — *June 2026 – Aug. 2026*

- Engineered fuel tank mounting structure and fuel filler line integration for Light Tactical Vehicle (LTV) platform using SolidWorks.
- Performed static structural FEA on tank mount under dynamic operational loads to evaluate structural integrity and ensure design compliance with safety factor requirements.
- Conducted CFD flow simulations across multi-bend fuel lines to evaluate pressure drop characteristics and optimize refueling throughput.

Aksantara ITB (Unmanned Aerial Vehicle Team) — Bandung, Indonesia
Head of Mechanical Design – Technology Development Division — *2025 – Present*

- Direct structural design and aeromechanical integration for competition hybrid fixed-wing and VTOL drones.
- Lead TUBITAK TEKNOFEST Turkey airframe team, optimizing composite and 3D printed structures for payload deployment.
- Supervised prototyping with composite carbon fiber layup, CNC machining, and additive manufacturing.

Vorione ITB (Renewable Energy Research Team) — Bandung, Indonesia
Mechanical Analyst & Designer — *2024 – Present*

- Developed CAD models and structural calculations for small-scale Vortex Shedding Turbines (VST) utilizing vortex-induced vibrations.
- Conducted dynamic stress and fatigue calculations on oscillating mast assemblies to optimize renewable power generation.

Projects & Research

Industrial Piping Network & Pig Receiver Design | *Autodesk Inventor, Stress Analysis, ASME* — 2025

- Engineered complete gas gathering piping system and pig receiver adhering to ASME B31.8 and API industrial standards.
- Conducted hydraulic sizing and FEA structural stress analysis, winning **1st Place** at MEDCOM UGM 2025.

Dimple Wing Aerodynamic Drag Reduction (ICARES 2024 Paper) | *ANSYS Fluent, CFD* — 2024

- Published research analyzing dimple surface configurations on small aircraft wings to reduce drag via vortex generation.
- Conducted multiphase CFD in ANSYS Fluent evaluating boundary layer behavior and wake turbulence reduction.

Tube-Launched Folding Wing UAV | *SolidWorks, 3D Printing, Machining* — 2025 – 2026

- Designed deployable wing kinematics for pneumatic container launch, maximizing post-launch span and structural stiffness.

Honors & Awards

1st Place – Kontes Robot Terbang Indonesia (KRTI), Technology Development (Airframe Project) — 2025
1st Place – National Piping Design Competition, MEDCOM Universitas Gadjah Mada — 2025
2nd Place – Kontes Robot Terbang Indonesia (KRTI), Technology Development (Airframe Project) — 2024

Technical Skills

CAD & 3D Modeling: SolidWorks, Autodesk Inventor, AutoCAD
Simulation & Analysis: ANSYS Fluent (CFD), FEA Stress Analysis, Flow Simulation, Hydraulic Sizing, Aspen HYSYS
Manufacturing & Prototyping: 3D Printing (Additive), CNC/Lathe/Milling Machining, Technical Drawing (GD&T)
Programming & Tools: Python, MATLAB, Microsoft Office, Google Workspace
Languages: Indonesian (Native), English (Professional Working Proficiency)